Neurosyphilis and HIV Coinfection

Of note, patients with cerebrospinal fluid (CSF) abnormalities should be cautiously diagnosed with asymptomatic neurosyphilis, as CSF leukocytosis or elevated protein levels occur in up to 60% of HIV-infected individuals not on effective antiretroviral therapy, even in the absence of syphilis.

Cardiovascular and Serologic Manifestations

The prevalence of cardiovascular syphilis may be increased in HIV-positive patients. One study from India reported a 14.3% prevalence among 14 HIV-infected individuals with syphilis, compared to 2% in 100 HIV-negative syphilis patients.

HIV-related dysregulation of the humoral immune response can lead to uncontrolled antibody production and markedly elevated titers. In one study, median rapid plasma reagin (RPR) titers were four times higher in HIV-infected patients (1:128) than in HIV-uninfected patients (1:32) during their first episode of early syphilis. B-cell activation also increases the rate of biologic false-positive nontreponemal tests. Conversely, delayed seroreactivity and false-negative nontreponemal tests have been observed, warranting a low threshold for biopsy of suspicious skin lesions. The sensitivity of treponemal tests may be reduced in HIV-infected patients previously treated for syphilis (80% vs. 97%).

Clinical Features in HIV Infection

Manifestations of secondary syphilis may be more common in HIV-infected individuals. Histopathologically, typical findings are usually present, although some cases may show a lymphoid infiltrate resembling cutaneous T-cell lymphoma.

Diagnostic Tests

Definitive diagnosis of early syphilis requires visualization of spirochetes in lesional exudate or tissue via darkfield microscopy or direct immunofluorescence. If diagnosis is not confirmed by these methods or by clinical evaluation of characteristic lesions with reactive serologic tests, skin biopsy is recommended for suspected primary, secondary, and especially tertiary syphilis lesions.

Histopathologic Examination

Chancre (Primary Syphilis)

The epidermis is acanthotic at the margins of the chancre and becomes thinner and edematous toward the eroded center. A dense dermal infiltrate composed predominantly of lymphocytes and plasma cells, with some histiocytes and a few neutrophils, is present centrally, with a perivascular distribution at the periphery. In untreated cases, *Treponema pallidum* can be visualized using silver stains such as Warthin-Starry, appearing gray. Spirochetes are most abundant in the epidermis and around dermal capillaries. Electron microscopy shows *T. pallidum* primarily in extracellular locations. Less commonly, organisms are found within endothelial cells of capillaries and inside vacuoles and phagolysosomes of plasma cells, macrophages, and neutrophils; occasionally within fibroblasts and nerve fibers.

Secondary Syphilis

Lesions typically show vascular dilatation and endothelial cell proliferation, with a superficial and deep perivascular dermal infiltrate rich in plasma cells. The infiltrate often has a lichenoid pattern, may obscure the dermoepidermal junction, and in about one-third of cases, surrounds adnexal structures and nerves.

- **Macular lesions**: Minimal or no epidermal changes.
- **Papulosquamous plaques**: Psoriasiform epidermal hyperplasia, parakeratosis, focal spongiosis, basal vacuolar changes, and microabscesses containing neutrophils.

Other common epidermal findings include epidermal pallor, keratinocyte necrosis, red blood cell extravasation, and mononuclear cell exocytosis. Dermal vessels exhibit endothelial proliferation and swelling. The inflammatory infiltrate is arranged in a "coat-sleeve" pattern around vessels and consists of lymphocytes, histiocytes, and numerous plasma cells. The pattern may be lichenoid, T-shaped, or purely perivascular, and may extend into the deep dermis. In approximately one-third of cases, the infiltrate surrounds adnexal, vascular, and neural structures.

Small epithelioid granulomas—sometimes containing Langhans or foreign-body-type giant cells—may be present, even in early secondary syphilis lesions.